3.5.5 Relationship Type

In ER Model, relationship exists between two entities. Three types of relationships are available and the Entity-Relationship(ER) diagram is based on the three types listed below.

One-to-One relationship: Consider two entities A and B. one-to-one (1:1) relationship is said to exist in a relational database design, if 0 or 1 instance of entity A is associated with 0 or 1 instance of entity B, and 0 or 1 instance of entity B is associated with 0 or 1 instance of entity A.

One-to-Many relationship: Consider two entities A and B. one-to-many (1:N) relationship is said to exist in a relational database design, for 1 instance of entity A there exists 0 or 1 or many instances of entity B, but for 1 instance of entity B there exists 0 or 1 instance of entity A.

Many-to-Many relationship:Consider two entities A and B. many-to-many (M:N) relationship is said to exist in a relational database design, for 1 instance of entity A there exists 0 or 1 or many instances of entity B, and for 1 instance of entity B there exists 0 or 1 or many instance of entity A.

In reality one-to-one are in less usage, where as one-to-many and many-to-many are commonly used. However in relational databases, many-to-many are converted into one-to-many relationships.

3.5.5.1 Relationship instance

It is a finite set of liples in the RDBMS systems relation instances never have duplicate . E.g if **Works-for** is the relationship between the Employee entity and the department entity, then Ram works for Comp.sc department, shyam works for electrical department ..etc are relationship instances of the relationship, works for.

3.5.5.2 Degree of a relationship

The number of entity types involved is known as Degree of relationship. One – Unary, Two – Binary, Three – Ternary.E.g An employee of an organization acts as manager of few other employees. It also connects one entity to itself as a loop. so **manager-of** is unary. Similarly employee **works-for**department, connects two entities and is binary. If a customer purchase an item, it involves shop keeper also and is a ternary relationship.

3.5.5.3 Cardinality

It is defined as the number of items that must be included in a relationship.ie number of entities in one set mapped with the number of entities of another set via the relationship. The three classifications in Cardinality are one-to-one, one-to-many and Many-to-Many. See Figure 3.9-3.11



Figure 3.9 Cardinality

In the above example we have two entities Person and Vehicle. If we consider a person driving vwchicle, then we have one-to-one relationship between Person and Vehicle. See Figure 3.10



Figure 3.10 CardinalityRelation 1 to n

In the above example, Customer places the Order is a one-to-many relationship.

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